

Gateway payment architecture

Introduction to the Gateway

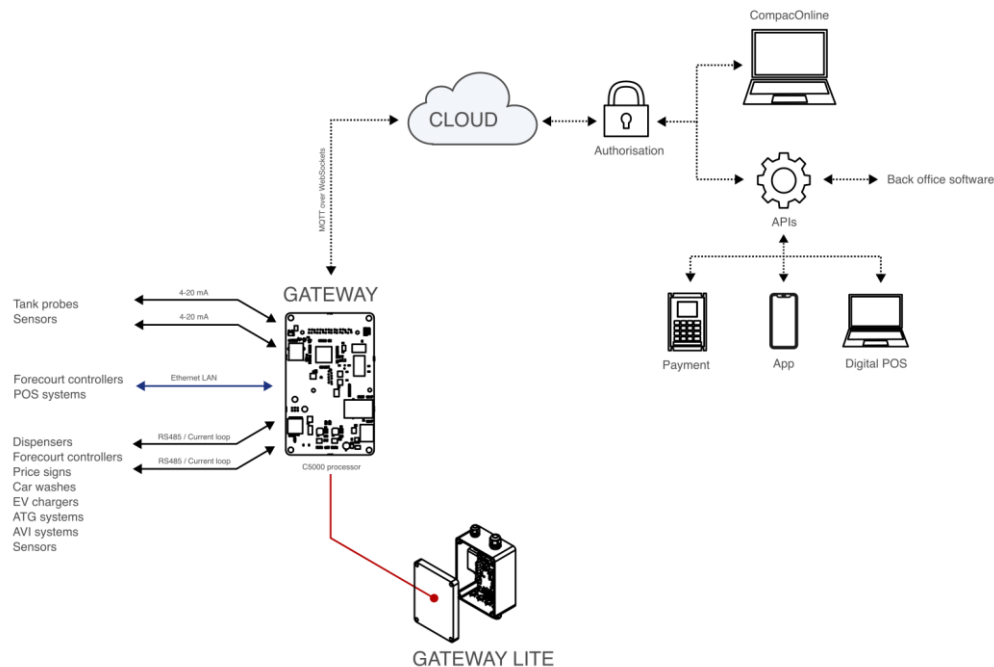
The Compac Gateway is a lightweight forecourt controller designed to act as a digitisation layer (IoT device) for an entire retail or commercial forecourt. The Gateway captures all forecourt protocol communication and connects all forecourt devices to a simple cloud (Azure) backend and exposes this information through a simple REST API on CompacOnline. For example:

- transactions (authorisation, completion, quantity, price),
- pump status,
- tank levels,
- price sign status,
- checking fuel cards, discount cards, prepay cards or Compac cards
- etc...

The Gateway also permits commands to be pushed to forecourt devices:

- reserving, authorising and cancelling pumps,
- changing prices (price signs and pumps),
- disabling devices
- etc...

The purpose of the Gateway is to enable an ecosystem of digital devices and services to have open and seamless access to all forecourt devices without the need to integrate with multiple vendors and integrates with all major forecourt controller brands.

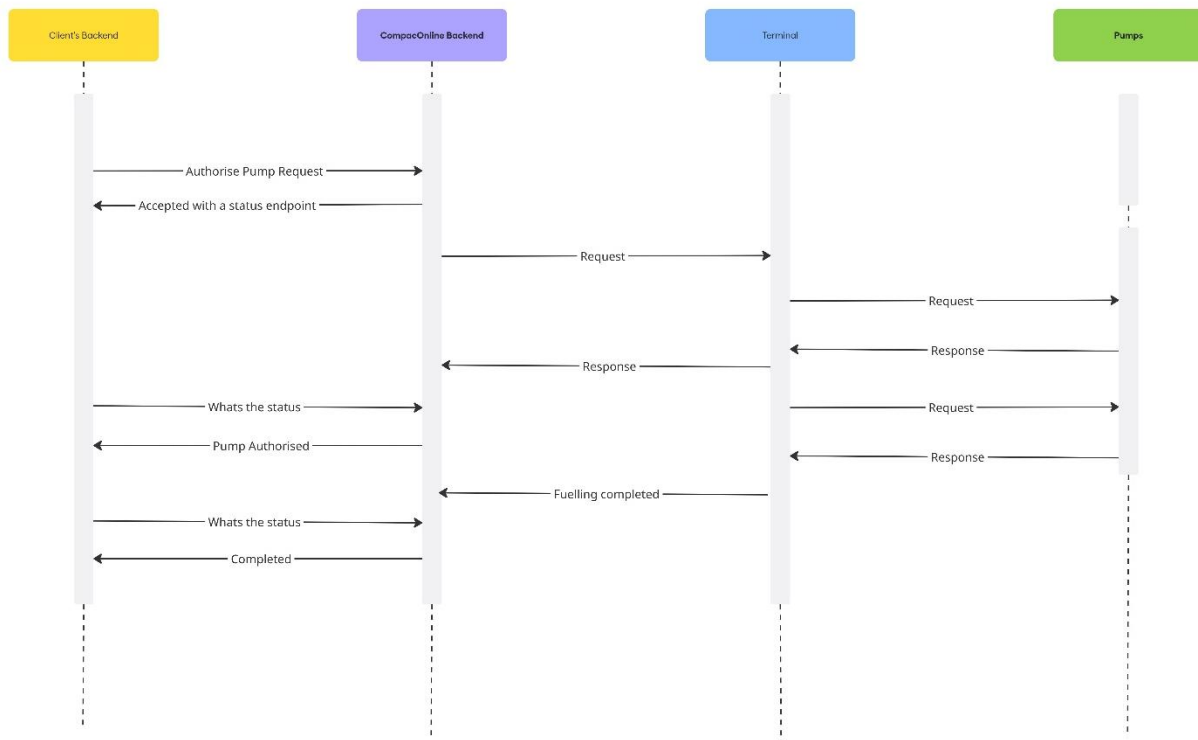


The Gateway is simple to install and cost effective enough to enable distributors to unify and digitise an entire network without changing the forecourt architecture at low cost.

Digital payments

The Gateway allows digital payments through the CompacOnline API with third party and Compac apps. Our proposal is to utilise this architecture to connect third party payment terminals to the forecourt through a backend-to-backend connection. The transaction flow can be summarised as:

- customer taps card at client payment terminal, selects pump (skipped if terminal configured as pay at pump), and enters pre-auth amount,
- client's backend server sends us an authorisation request with pump number and amount, and we reply with 202 Accepted with a unique id,
- CompacOnline backend sends a request to the Gateway on site to authorise the pump,
- gateway communicates with pump as usual,
- gateway sends a reply to the backend and backend saves it,
- client polls the status endpoint to check for status,
- when the fuelling finishes, Gateway sends a message to the backend,
- when client polls us next time, our backend returns completed,



Discussion

The proposed architecture allows a POS or payments company to provide a lightweight and flexible in store, forecourt, or pay at pump payment solution requiring a single simple integration with the CompacOnline API. The proposed architecture simplifies the installation of onsite payment hardware to simply an internet connection and requires next to no configuration.